

Task-Aware Reuse of Cellular Molecular Receivers: Response Prediction and Feedback-Driven Waiting

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Abstract—A living cell used as a molecular receiver keeps a memory of earlier commands, so the next command can fail if it is sent too early. We ask what information predicts a cell’s response to the next command and whether feedback can replace a fixed guard interval. On experimental FGF2–ERK trajectories of 1,565 single cells, nested predictors show that causal filtering of the reporter and the increment of a calibrated mechanistic (B3) model reduce the 10-min forecast error by 9.1% and 4.5% (18.8% and 9.8% once the reporter noise floor is removed), whereas explicit input history adds at most 1%; just before a new command, however, the mechanistic increment raises the error by 7.2%, because B3 overestimates how far the receiver has recovered. In B3 simulations with a 90% reliability constraint for each previous command, feedback-driven waiting closes 42–45% of the gap between history-conditioned fixed waits and an oracle (7–8 min). With a feedback delay, the rule must predict readiness ahead; the mechanistic model then pays off at high reporter noise or sparse sampling, while a smoothed reporter with its trend suffices otherwise.

Index Terms—Molecular communication, cellular receivers, mechanism-informed prediction, Bayesian filtering, receiver reuse, waiting rules, FGF2, ERK.

I. INTRODUCTION

Molecular communication (MC) with living cells as receivers offers a natural interface between engineered communication systems and biological processes. Unlike a conventional receiver, a cell retains the effect of earlier chemical stimuli in receptor dynamics, intracellular signaling and feedback, so repeated commands are not independent: a receiver that has just responded may not execute the next command with the same reliability. This raises a communication-level question for repeated molecular tasks: *when can a living receiver be used again, and what information does that decision need?*

Receiver memory is familiar in MC, but it is usually placed in the channel. Diffusion leaves residual molecules that cause intersymbol interference (ISI) [1], [2], and receivers counter it with sequence detection and equalization [3], decoders that exploit memory and sampling rate [4], adaptive thresholds [5], and symbol or detection intervals that trade rate against residual ISI [6], [7]; ligand-binding reception adds its own noise [8]. In these designs the guard interval is set in advance. For a cellular receiver the memory is a hidden biochemical state seen only through a noisy reporter, and deciding from that reporter when to send the next command is a stopping problem, related to sequential detection [9], [10] and optimal stopping [11]. History-dependent FGF2–ERK responses [12], frequency-dependent signaling [13] and feedback control

of single cells [14], [15] show that such memory exists, and experimental biological receivers are emerging [16]–[19]. What is missing is a measurement-based account of which information predicts a cell’s response to the next command, and a test of whether feedback can replace a fixed guard interval.

We combine experimental FGF2–ERK single-cell trajectories with the calibrated B3 signaling model of [12]. Nested predictors add the current reporter, causal reporter filters, lagged values, a mechanistic B3 increment and explicit input history one at a time. Receiver reuse is posed as a calibrated stopping rule on B3 with a reliability constraint for each previous command; a feedback delay links the two parts, because with delay the rule must predict readiness ahead, as the forecasts do. Our contributions are:

- On measured cells, reporter filtering and the B3 increment carry the forecast gain (15.5% noise-adjusted skill at 10 min, 21% at 20–30 min), while input history adds at most 1% and destabilizes longer forecasts; the B3 increment fails just before a new command, and natural probes in the data show why: B3 overestimates recovery after a long command.
- Under a per-history 90% reliability constraint, feedback-driven waiting closes 42–45% of the gap between history-conditioned fixed waits and the oracle; a smoothed reporter is as good as the B3 belief at the reference noise.
- A condition map over feedback delay, sampling interval and noise shows when a mechanistic model is needed: readiness must be predicted rather than nowcast once the delay reaches about 20 min, and the B3-predictive rule beats a smoothed reporter with its trend by 0.9–5.5 min at high reporter noise and by up to 1.1 min with sparse sampling, but not for clean, dense feedback.

II. SYSTEM AND TASK MODEL

Let $u(t) \geq 0$ be the commanded FGF2 concentration (ng/ml). Delivery to the receiver is represented abstractly as

$$\begin{aligned}\tilde{u}(t) &= \mathcal{T}_\theta[u](t), & \dot{x}_j &= f_{\theta_j}(x_j, \tilde{u}), \\ z_j &= g(x_j), & y_{j,i} &= z_j(t_i) + \nu_{j,i},\end{aligned}\tag{1}$$

where $\tilde{u}$ is the local concentration, x_j the hidden state of cell j , z_j its noise-free normalized ERK reporter and $y_{j,i}$ the observation at frame t_i with noise $\nu_{j,i}$. Neither $\mathcal{T}_\theta$ nor $\tilde{u}$ is calibrated here. The information $\mathcal{H}_t$ holds the executed commands and the observations that have arrived by t .

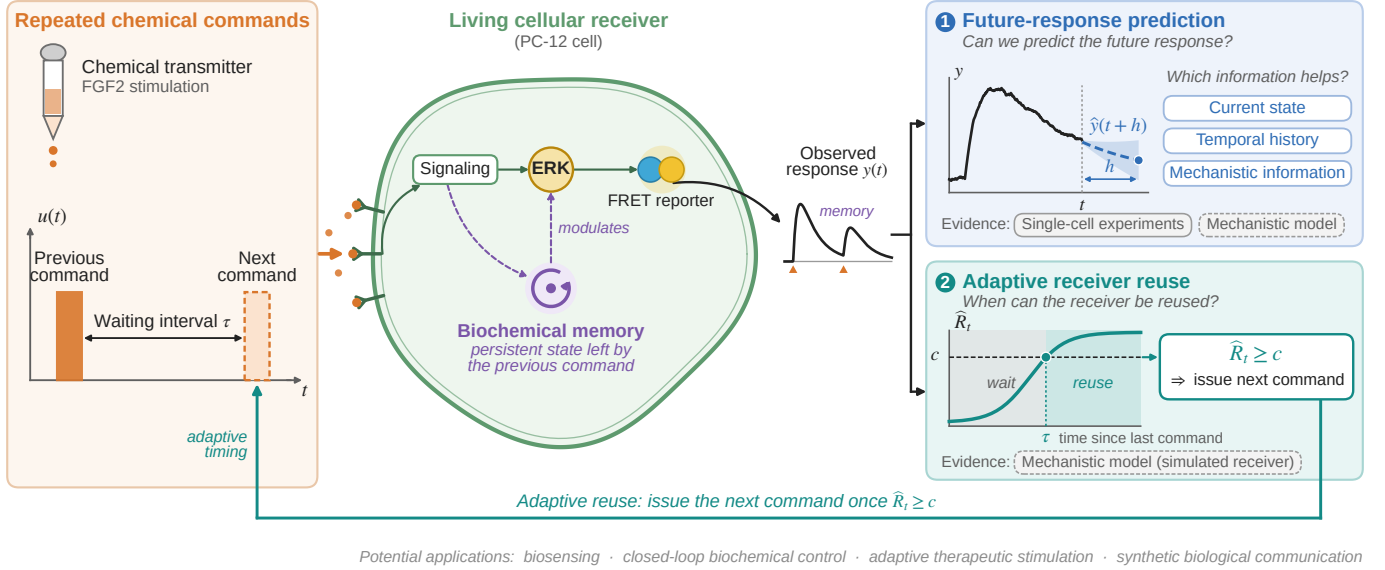


Fig. 1. System concept. Repeated FGF2 commands drive a living receiver whose biochemical memory shapes the ERK/FRET response $y(t)$ to later commands. The response supports (1) prediction of $\hat{y}(t+h)$ and (2) readiness-based reuse, where the next command is sent once $\hat{R}_t \geq c$.

Reuse task. The amplitude and duration of the next command are fixed; only its timing is decided. A previous command of $m \in \{3, 30\}$ min ends at time 0, and a 5-min probe at the same amplitude acts on the receiver at $\tau \in \{0, 2, \dots, 120\}$ min. With z_j^τ the response when the probe acts at τ and r_j^{nv} the same receiver's rise to the probe without a previous command,

$$S_j(\tau) = \mathbf{1} \left\{ \max_{s \in [\tau, \tau+W]} z_j^\tau(s) - z_j^\tau(\tau) \geq \kappa r_j^{\text{nv}} \right\}, \quad (2)$$

$$R_{j,t} = \mathbb{P}(S_j(t) = 1 \mid \mathcal{H}_t),$$

with window $W = 20$ min. Success is scored on the noise-free response; decisions see only noisy observations. The completion time is $T = \tau + W$ on success and $T_f = 140$ min on failure, the latest completion of any success. The design problem is

$$\min \mathbb{E}[T \mid m] \quad \text{s.t.} \quad \mathbb{P}(S = 1 \mid m) \geq q, \quad m \in \{3, 30\}, \quad (3)$$

with $q = 0.9$ an empirical calibration target, not a coverage guarantee. Requiring q for each previous command, not on average, prevents a rule from buying speed after one history with failures after the other. We use $\kappa = 0.5$, the largest value on $\{0.30, 0.35, \dots, 0.70\}$ at which the oracle still succeeds for $> 99\%$ of simulated receivers within 120 min (99.25%; 95.1% at 0.55, 82.0% at 0.60). If the receiver's state reaches the controller late, or the command takes time to arrive, a probe decided at t acts at $t+d$: the rule must predict R at $t+d$ from $\mathcal{H}_t$, a forecast with horizon $h = d$.

III. PREDICTORS AND WAITING RULES

A. Nested Predictors

Predictors for horizons $h \in \{2, 10, 20, 30\}$ min start after a fixed 10-min warmup; future reporters are targets only. Table I

TABLE I
NESTED PREDICTORS (CODE KEYS IN THE REPOSITORY: P, C, O, A, M, A+M, H). ALL LEARNED PREDICTORS KNOW THE FUTURE COMMAND; RIDGE REGRESSION WITH EQUAL CONDITION WEIGHTS.

Predictor	u_t, y_t, slope	$\bar{y}_\ell$	lags	$\hat{\delta}_{t,h}$	$v_\ell, \int u$
Persist ($\hat{y}_{t+h} = y_t$)					
Current	✓				
Filter	✓	✓			
Lag	✓		✓		
Filter+B3	✓	✓		✓	
Lag+B3	✓		✓	✓	
F+B3+Input	✓	✓		✓	✓

lists the feature sets; lags are the reporter and command values of the previous 5–30 frames. The causal reporter and input filters with scales $\ell \in \{2, 10, 60\}$ min are

$$\bar{y}_{\ell,i} = e^{-\Delta_i/\ell} \bar{y}_{\ell,i-1} + (1 - e^{-\Delta_i/\ell}) y_i, \quad (4)$$

$$v_\ell(t) = \frac{1}{\ell} \int_0^t e^{-(t-s)/\ell} u(s) ds,$$

with $\bar{y}_{\ell,0} = y_0$ and $\Delta_i = t_i - t_{i-1}$. As mechanism we execute the authors' B3 model with their 1,000 posterior samples. With $f_s(t)$ the noise-free reporter of sample s under the recorded command, the samples are weighted by the cell's past reporter,

$$w_s(t) \propto \exp \left[- \sum_{t_i \leq t} (y_i - f_s(t_i))^2 / (2\sigma^2) \right], \quad (5)$$

$$\hat{\delta}_{t,h} = \sum_s w_s(t) [f_s(t+h) - f_s(t)],$$

a fixed-parameter particle approximation of the B3 belief that uses the same information as the learned predictors. Since $\hat{\delta}_{t,h}$ already uses the command history, F+B3+Input tests only the extra value of explicit input features. Penalty, σ and lag length are chosen on validation data; normalization, lag filling and fitting details are in the repository.

B. Calibrated Waiting Rules

A waiting rule probes at the first decision time at which its readiness estimate reaches a threshold c , or at 120 min,

$$\tau = \min(\{t : \hat{R}_t \geq c\} \cup \{120\}), \quad (6)$$

a stopping time on the observations, as in sequential tests [9]. Three estimators of R_t are compared on B3: *current* uses only y_t , through a Gaussian kernel over the (reporter, success) states of reference receivers; *smoothed* applies the same kernel to a causal exponential smoothing of the observations (scale 2–16 min); the *B3 belief* weights reference receivers under the known command history by all observations, as in Eq. (5), with likelihood s.d. equal to the noise s.d. floored at 0.001. The comparator is one fixed wait per previous command, a history-conditioned guard interval. References are one half of the posterior samples (calibration; each receiver left out of its own references); the other half (evaluation) never enters references, thresholds or waits. Every rule, fixed or adaptive, chooses one parameter per previous command that minimizes that history’s calibration completion subject to its calibration success $\geq$ a target, i.e. Eq. (3) on calibration samples. Twenty splits redraw the halves and the observation noise.

With delay d , the *B3-predictive* rule averages the references’ success at $t + d$ under the same weights, $\hat{R}_{t+d|t} = \sum_s w_s(t) S_s(t + d)$; the *B3 nowcast* ignores the delay. The model-free comparator also extrapolates: a kernel over the smoothed level and its last increment, $(\bar{y}_t, \bar{y}_t - \bar{y}_{t-\Delta})$, with the covariance of smoothed white noise, maps the current trend to success at $t + d$. Observations arrive every $\Delta \in \{2, 6, 10\}$ min and decisions are made when they arrive.

IV. DATA AND EVALUATION

All numerical files come from the paper-linked repository at a pinned commit [20], [21]; commands follow the authors’ model XML, and per-cell normalization uses only frames before stimulation onset. Table II lists the 22 conditions. Roles were frozen before any newly used response file was downloaded: Test A is a new protocol, Test B’s timing was inferred from the authors’ simulations, and Test C uses new concentrations recorded in the session of inspected mixed blocks. Because the authors fitted B3 to the sustained and 3/20 means and compared architectures on the single 5-min and mixed blocks at 2.5 and 250 ng/ml, B3 is a calibrated reference, not a model blind to those protocols; it is executed from the authors’ files without refitting (posterior-predictive mean within 8.5×10^{-6} of their exports). In the corrected leave-one-protocol-out (LOPO) each of six protocol groups is held out with predictor tuning restricted to the training protocols. Forecast rows are assigned to the phases stimulation, next command (a pulse starts in $(t, t + h]$), early washout (≤ 30 min after a pulse) and late. The score is the square root of the equally weighted condition MSE,

$$\text{RMSE}(h) = \left[\frac{1}{|\mathcal{K}|} \sum_{k \in \mathcal{K}} \frac{1}{N_{k,h}} \sum_{i \in k} (\hat{y}_{i+h} - y_{i+h})^2 \right]^{1/2}, \quad (7)$$

TABLE II
PROTOCOLS, ROLES AND SINGLE-CELL TRAJECTORIES (22 CONDITIONS, 1,565 CELLS, 2-MIN FRAMES). B3 USE: CONCENTRATIONS WHOSE MEANS THE AUTHORS FITTED (FIT) OR USED IN THEIR ARCHITECTURE COMPARISON (ARCH.).

Protocol	Role	ng/ml	Cells	B3 use
Sustained	Train	0.25, 2.5, 25, 250	248	fit 2.5, 250
3/20 pulses	Train	2.5, 25, 250	207	fit 2.5, 250
Single 5 min	Val.	0.25, 2.5, 25, 250	370	arch. 2.5, 250
Single 10 min	Test A	0.25, 2.5, 25, 250	248	–
Single 60 min	Test B	2.5, 25, 250	167	–
Mixed	Test C	0.25, 25	171	–
Mixed	Viewed	2.5, 250	154	arch. 2.5, 250

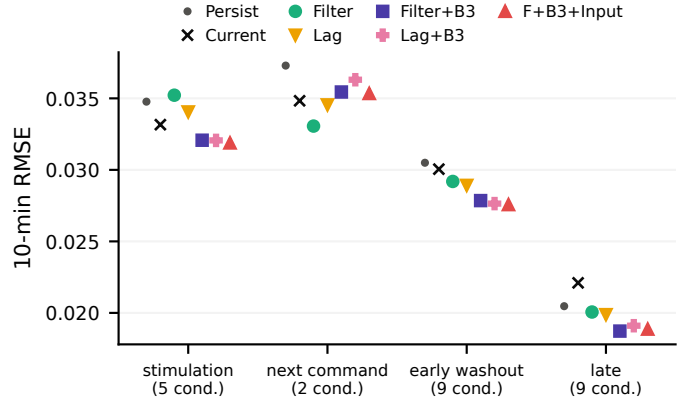


Fig. 2. Condition-equal 10-min RMSE on Tests A–C by forecast phase (axis not starting at zero).

over the conditions $\mathcal{K}$ of a role. Because every error contains the reporter noise, we also report a noise-adjusted skill, $1 - (\text{MSE} - \hat{\sigma}_v^2) / (\text{MSE}_{\text{Persist}} - \hat{\sigma}_v^2)$, with $\hat{\sigma}_v$ a robust second-difference estimate of white reporter noise per condition (0.006–0.009), and 95% intervals from resampling cells within conditions. Cells share a session and no day identifiers exist, so intervals describe cell sampling, not independent experiments. Every analysis was frozen in its own protocol before running, but all are post hoc with respect to inspected data.

V. RESULTS AND DISCUSSION

A. Which Information Predicts the Response

Table III(a) reports the nested predictors. On Test A at 10 min, reporter filtering reduced the error by 9.1% relative to *Current*, the B3 increment by 4.5% relative to *Filter*, and input history by 1.0% relative to *Filter+B3*, giving 0.02098, 8.2% below *Persist*; these reductions use different baselines and are not additive. Removing the white-noise floor, 8.9% of the *Persist* MSE at 10 min, roughly doubles these shares to 18.8%, 9.8% and 2.2%, and *Filter+B3* explains 15.5% of the noise-free *Persist* error (Table III(c)); the cell-bootstrap intervals of the three RMSE reductions were 7.4–10.9%, 3.0–6.1% and 0.6–1.4%. The input-history gain was small and not consistent across conditions (two of four on Test A; 0.4% and 0.1% on B and C). The lagged baseline *Lag* was 2.3% worse than *Filter* on Test A, but adding the B3 increment

TABLE III

CONDITION-EQUAL 10-MIN RMSE UNLESS NOTED. (A) VAL.: SINGLE 5-MIN PULSE; A: SINGLE 10-MIN PULSE; B: SINGLE 60-MIN PULSE; C: MIXED PULSES AT 0.25 AND 25 NG/ML. BOTTOM ROWS COUNT CONDITIONS WITH LOWER MSE. (B) CORRECTED LOPO; B3 USE AS IN TABLE II ($^{\circ}$: OTHER CONCENTRATIONS). (C) NOISE-ADJUSTED SKILL (%) VERSUS PERSIST ON TEST A WITH 95% CELL-BOOTSTRAP INTERVALS. BOLD: SMALLEST PER COLUMN (A) OR ROW (B).

(a) Fixed roles					
Model	Val.	A	B	C	A, 2 min
Persist	0.02353	0.02287	0.02651	0.02876	0.01302
Current	0.02503	0.02442	0.02715	0.02825	0.01342
Filter	0.02403	0.02220	0.02715	0.02827	0.01283
Lag	0.02343	0.02272	0.02565	0.02769	0.01300
Filter+B3	0.02197	0.02120	0.02448	0.02658	0.01263
Lag+B3	0.02161	0.02114	0.02474	0.02652	0.01270
F+B3+Input	0.02195	0.02098	0.02439	0.02656	0.01260
Filter < Current	4/4	4/4	2/3	1/2	4/4
Filter+B3 < Filter	4/4	3/4	2/3	2/2	3/4
Lag+B3 < Lag	4/4	4/4	3/3	1/2	4/4
F+B3+Input < Filter+B3	1/4	2/4	2/3	1/2	2/4

(b) Leave-one-protocol-out						
Held-out	Filter	F+B3	F+B3+In	Lag	Lag+B3	B3 use
Sustained	0.02102	0.01969	0.01941	0.02025	0.01931	fit
3/20 pulses	0.03680	0.02886	0.02830	0.03287	0.02837	fit
Single 5 min	0.02243	0.02153	0.02139	0.02196	0.02134	arch.
Single 10 min	0.02084	0.02106	0.02104	0.02116	0.02082	none
Single 60 min	0.02701	0.02410	0.02369	0.02486	0.02422	none
Mixed, 0.25/25	0.02709	0.02623	0.02610	0.02706	0.02628	arch. $^{\circ}$

(c) Noise-adjusted skill by horizon (Test A)

Horizon h (min)	2	10	20	30
Filter	4.0	6.3	12.7	9.0
Filter+B3	8.2	15.5	21.1	21.0
95% interval	6–11	11–19	18–24	17–25
F+B3+Input	8.8	17.4	–1.7	13.5
F+B3 (phase)	3.6	14.8	23.4	23.4
Crossing BA, Filter / F+B3	–	56 / 55	52 / 57	53 / 58

to it still reduced the error by 6.9% (four of four concentrations), so the mechanistic gain does not depend on how the history is represented. In the corrected LOPO (Table III(b)), F+B3+Input beat Filter+B3 in all six folds (by 0.1–2.0%) and Lag+B3 beat Lag in all six, but with the 10-min pulse held out, one of the two protocols unseen by B3, Filter+B3 was 1.1% worse than Filter. At longer horizons the mechanistic increment mattered more: on Test A its skill rose to 21% at 20 and 30 min against 9–13% for Filter, and it anticipated the half-decay crossing of the reporter, the event a delayed reuse decision needs, more often (balanced accuracy 57–58% versus 52–53%; 65–67% versus 54–57% on Test B), although no predictor did so reliably. Explicit input history became harmful: F+B3+Input fell to –1.7% skill at 20 min on Test A and to –50% and –53% on Test B at 20 and 30 min.

B. The Next-Command Window

By phase (Fig. 2), the B3 increment reduced the error of Filter during stimulation (8.9%), early washout (4.6%) and late washout (6.6%), but raised it in the next-command window

TABLE IV

WAITING ON B3 (PRIMARY SPLIT, NOISE S.D. 0.005 UNLESS NOTED) WITH ONE PARAMETER PER PREVIOUS COMMAND, CALIBRATED TO 0.94, THE SMALLEST PRE-SPECIFIED TARGET AT WHICH EVERY RULE MET 0.9 FOR BOTH HISTORIES IN ALL 20 SPLITS AT NOISE 0.005. S_m (%), T_m (MIN): EVALUATION SUCCESS AND MEAN COMPLETION AFTER THE m -MIN COMMAND. ΔT : COMPLETION MINUS THAT OF THE FIXED WAITS; G : SHARE OF THE FIXED-TO-ORACLE GAP CLOSED (BOTH SPLIT MEDIANS); MET: SPLITS WITH $S_3, S_{30} \geq 90\%$.

Rule	S_3	T_3	S_{30}	T_{30}	ΔT	G (%)	Met
Fixed per history (94/110)	95.0	115.3	95.6	130.4	–	–	20/20
Current obs.	95.0	116.5	95.4	128.2	+0.2	–1	20/20
Smoothed	94.0	108.4	94.6	119.6	–7.7	45	20/20
B3 belief	95.4	109.0	95.0	119.9	–7.2	42	20/20
Oracle	100.0	99.0	98.4	110.4	–	100	–
Noise s.d. 0.01 (same fixed waits)							
Smoothed	94.4	111.8	93.6	122.5	–4.4	26	20/20
B3 belief	95.4	110.5	94.4	121.1	–6.0	35	19/20

from 0.03306 to 0.03544. That window holds two mixed conditions with ten forecast origins each: the increment helped at 0.25 ng/ml (0.03652 to 0.03485) and hurt at 25 ng/ml (0.02920 to 0.03603). Two phase-gated variants fitted on the training protocols did not repair this: a switch to Filter in next-command rows was rejected by the training data, where the increment helped (in-sample MSE 6.3×10^{-4} versus 8.4×10^{-4}), and phase-specific weights raised the next-command error to 0.03571 (held-out mixed fold: 0.03630, versus 0.03537 for Filter+B3 and 0.03302 for Filter). The training protocols contain no probe after a long command, so their next-command windows cannot teach the correction. Natural probes in the measured data explain the failure (Sec. V-E).

C. Waiting under a Per-History Reliability Constraint

Readiness and waiting were simulated on B3 with 25 ng/ml commands and observations every 2 min. Readiness depended almost only on the current reporter, waits of 6–48 min never succeeded, and no state met the criterion without the probe. In Fig. 3(a,b) every reliability–latency frontier lies between the fixed waits and the oracle, and the smoothed and B3-belief frontiers nearly coincide. Calibrated to 0.90, every rule, fixed waits included, met 0.9 for both histories in only 7–11 of 20 splits, so the per-history constraint is tight at the edge of the target. At 0.94 all rules met it in every split (Table IV): feedback finished 7.2 min (B3 belief) and 7.7 min (smoothed) earlier than per-history fixed waits of 94 and 110 min, faster in all 20 splits, and closed 42% and 45% of the gap to the oracle (38–42% at targets 0.90 and 0.92). The current-reporter rule closed none of it. Under the pooled constraint of our earlier analysis, the B3 belief and smoothing closed 33% and 38% of the gap to pooled history-conditioned waits (4.8 and 5.5 min). At noise s.d. 0.01 the B3 belief beat smoothing in all splits (by 1.5–1.7 min; 35% versus 26% of the gap), whereas at 0.005 smoothing was 0.1–0.4 min faster. Larger failure penalties ($T_f = 160$ –260 min) raised the gain over fixed waits from 5.5 to 5.9–8.7 min with the same order of rules. Earlier sensitivity analyses (repository) agree: the gain persists at $\kappa = 0.4$ and

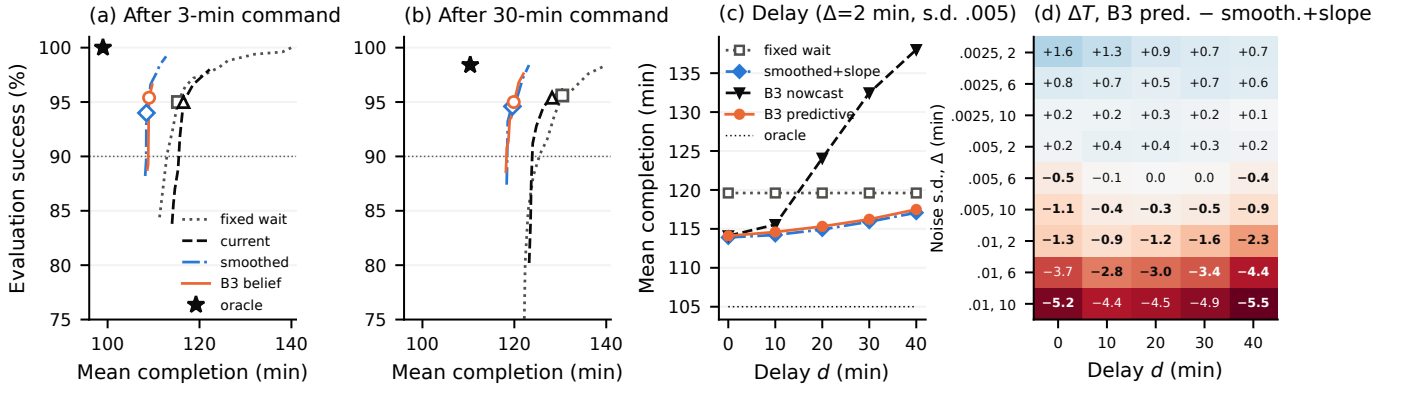


Fig. 3. Waiting on the authors' B3 model (25 ng/ml; task: probe rise $\geq 50\%$ of the naive rise within 20 min). (a,b) Reliability–latency frontiers on evaluation samples (primary split, noise s.d. 0.005; thresholds or waits swept, descriptive), with the operating points calibrated to 0.94 (open markers). (c) Median completion over 20 splits versus feedback delay d (calibration target 0.90 per history). (d) Condition map: median completion of the B3-predictive minus the smoothed+slope rule (min; negative: the model is faster) over delay, noise s.d. and sampling interval Δ ; bold: model faster in ≥ 16 of 20 splits at no more than 0.5 points lower success. One simulated model structure, not measured cells.

at a shared calibration margin, at $\kappa = 0.6$ even the oracle fails for 18.5% of receivers, and under correlated noise the rules need a likelihood that models the correlation (82.0% success otherwise, 94.5% with an AR(1) likelihood).

D. Feedback Delay: When the Mechanism Pays Off

With delay, readiness must be predicted: the B3 nowcast became slower than the fixed waits from $d = 20$ min on (124–138 min), whereas both predictive rules lost only about 3 min between $d = 0$ and 40 min (Fig. 3(c)), so the value of feedback over fixed waits shrank from 5.5–5.9 to 2.2–2.6 min. The condition map (Fig. 3(d)) answers when the model is needed. At noise s.d. 0.01 the B3-predictive rule was faster than smoothing plus slope in every split of every cell, by 0.9–5.5 min, and with sampling every 6 or 10 min the model-free rule became slower than the fixed waits (by 0.5–4.7 min) while the B3 rule still beat them. At the reference noise the model helped with 10-min sampling (0.3–1.1 min) and in two of five delays with 6-min sampling; with 2-min sampling and at noise 0.0025 the smoothed trend was 0.1–1.7 min faster, and at low noise the B3 belief lost reliability as its weights collapsed (median success 86.5–88.0% with 2-min sampling). The model was faster in at least 16 of 20 splits in 18 of the 45 cells; four noise-0.01 cells miss the class only because the B3 rule's median success was 0.6–1.7 points lower.

E. Natural Probes in Measured Cells

Two recorded protocols contain later pulses that act as probes at fixed waits (25 ng/ml): in the 3/20 protocol each 3-min pulse comes 20 min after the previous one, with the cell's own first-pulse rise as naive reference (58 cells); in the mixed protocol a 5-min pulse comes 60 min after a 30-min pulse, scored against single 5-min pulse cells (84 cells). Both probes were early by B3's standard: B3 predicted no success for any sample, and every estimator called no cell ready. The measured cells agreed: 19% met the criterion at the first 3/20 probe and 3–9% at later ones, and none at the mixed probe,

so the rules made no false ready call but missed the 19%. The rises differed, however: at the mixed probe measured cells reached a median 4% of the naive rise where B3 predicts 31%, i.e. B3 overestimates recovery after a long command, the same error that raised the next-command forecast error at 25 ng/ml. At the first 3/20 probe B3 underestimated the rise (8% versus 16% of the naive rise). The B3 weights collapsed on measured cells (effective sample size 1.0–1.4), so a real-data filter needs process noise or cell heterogeneity, and the measured noise floor (0.007–0.008) lies where the map favors model-based filtering only if the model itself is right.

F. Implications for MC Design

A previous command acts on the next one like ISI whose memory sits in the receiver's biochemistry rather than only in the channel [2], [6]. A fixed wait is a guard interval; conditioning it on the known previous command is the first step, and requiring reliability for each previous command lengthens the guard after the long command (110 versus 94 min). Feedback then adapts to state variability within each history, much as adaptive detection adapts to residual interference [4], [5], and recovers about 40% of the gap to the oracle. Which estimator feedback needs depends on the noise, the sampling interval and the loop delay: a smoothed reporter with its trend suffices for frequent, clean feedback, whereas a model is worth its cost for noisy or sparse feedback, provided it is right about recovery; with a loop delay, either estimate must be predictive. A prospective test should randomize 3- versus 30-min preconditioning per chamber, probe at 60–120 min, measure the naive response in same-session controls and count sessions rather than cells.

G. Evidence Boundaries

Data. The prediction comparisons are post hoc on inspected roles; in the frozen analysis of the main fresh test, history features alone improved on persistence by only 0.7%; Test C shares a session with inspected data, no experimental-day identifiers exist, protocol generalization is therefore not cross-day

generalization, and B3 had seen some protocols; the noise floor assumes white reporter noise. *Model*. Waiting results come from one simulated model structure whose posterior samples express parameter uncertainty, not validated cell heterogeneity; the 20 splits reuse one finite posterior set; the B3 rules share the plant's structure, which favors them; the targets have no coverage guarantee and a calibration margin is not one. *Task*. $\kappa = 0.5$ was adopted after an 80% version proved infeasible; the naive response is a simulation counterfactual; amplitude and duration were fixed; the natural probes cover one dose at two fixed waits and score readiness calls, not completion time; the experimental predictors were not tested as decision models, so forecast gains do not imply better waiting decisions.

VI. CONCLUSION

On experimental FGF2–ERK trajectories, reporter filtering and a fixed calibrated mechanistic increment carry the forecast gain while explicit input history adds little, but the mechanistic increment fails just before a new command because B3 overestimates recovery after a long command, as natural probes in the data confirm. In B3 simulations with a reliability constraint for each previous command, feedback-driven waiting closes about 40% of the gap between history-conditioned guard intervals and the oracle; a smoothed reporter suffices for frequent, clean feedback, a mechanistic model pays off when the feedback is noisy or sparse, and any feedback delay must be predicted rather than ignored.

DATA AND CODE AVAILABILITY

Code, frozen results, protocols with hash records, implementation details and verification scripts are available at <https://github.com/ruifeng105/task-aware-cellular-mc>; one command re-runs the analysis and checks the reported values against a frozen reference. Sources are credited in [12], [20], [21]; author files keep their attribution and receive no new license.

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